SUBCHAPTER 3.4 · ROTATIONAL DYNAMICS & TURBOMACHINERY

Conservation of Angular Momentum & Turbomachinery Torque

Moment balance about strategic reduction centers $\vec{x}_0$, elimination of unknown reactions, and Euler's fundamental turbomachinery equation.

1. The Master Angular Momentum Equation

In continuum mechanics, the moment of momentum (angular momentum) of a closed material fluid system $V_f(t)$ about an arbitrary fixed point in space $\vec{x}_0$ changes at a rate equal to the resultant external moment acting on the fluid:

$$\frac{d}{dt}\left[\int_{V_f(t)} \rho[(\vec{x}-\vec{x}_0)\times\vec{v}]\,dV\right] = \sum \vec{M}_{\vec{x}_0,\text{ext}}$$

[Eq. 3.37] Angular momentum principle for a material fluid volume.

Substituting intensive angular momentum $\vec{b} = (\vec{x}-\vec{x}_0)\times\vec{v}$ into the universal Reynolds Transport Theorem yields the **Master Integral Angular Momentum Equation** for any moving control volume $V_c(t)$:

$$\mathbf{\frac{d}{dt}}\left[\int_{\mathbf{V_c(t)}} \rho[(\vec{\mathbf{x}}-\vec{\mathbf{x}}_\mathbf{0})\times\vec{\mathbf{v}}]\,\mathbf{dV}\right] + \int_{\mathbf{\Sigma_c(t)}} \rho[(\vec{\mathbf{x}}-\vec{\mathbf{x}}_\mathbf{0})\times\vec{\mathbf{v}}][(\vec{\mathbf{v}}-\vec{\mathbf{v}}_\mathbf{c})\cdot\vec{\mathbf{n}}]\,\mathbf{d\sigma} = \sum \vec{\mathbf{M}}_{\vec{\mathbf{x}}_\mathbf{0},\text{ext}}$$

[Eq. 3.38] Universal Integral Angular Momentum Conservation Equation.

The external moments $\sum \vec{M}_{\vec{x}_0,\text{ext}}$ consist of pressure forces, viscous shear forces, body forces (e.g., gravity), and mechanical holding torques applied by shaft couplings or mounting flanges:

$$\sum \vec{M}_{\vec{x}_0,\text{ext}} = -\int_{\Sigma_c} (\vec{x}-\vec{x}_0)\times(p\vec{n})\,d\sigma + \int_{\Sigma_c} (\vec{x}-\vec{x}_0)\times(\bar{\bar{\tau}}'\cdot\vec{n})\,d\sigma + \int_{V_c} \rho(\vec{x}-\vec{x}_0)\times\vec{f}_m\,dV + \vec{\mathcal{M}}_{\text{shaft}}$$

Angular Momentum: Complete Derivation Chain (Notes.pdf, Eqs. 3.37-3.38)

Step 1: Angular momentum of a fluid volume (Eq. 3.37)

Let $\vec{r} = \vec{x} - \vec{x}_0$, with $\vec{x}_0$ a fixed point of an inertial frame. A fluid particle of mass $\rho\, dV$ has angular momentum $\vec{r} \wedge \rho\vec{v}\, dV$ about $\vec{x}_0$ (Notes, p. 33), so the angular momentum of the fluid volume is $\int_{V_f} \rho\, \vec{r} \wedge \vec{v}\, dV$. Newton's second law for angular momentum states that its rate of change equals the moment of the external forces about $\vec{x}_0$. The moment of the surface force $\bar{\bar{\tau}} \cdot \vec{n}\, d\sigma$ on each element of the boundary and the moment of the body force $\rho \vec{f}_m\, dV$ on each particle give Eq. 3.37:

$$\frac{d}{dt} \int_{V_f(t)} \rho\, \vec{r} \wedge \vec{v}\, dV = \int_{\Sigma_f(t)} \vec{r} \wedge (\bar{\bar{\tau}} \cdot \vec{n})\, d\sigma + \int_{V_f(t)} \rho\, \vec{r} \wedge \vec{f}_m\, dV$$

Eq. 3.37

Step 2: Consistency with the local momentum equation (extension)

Extension, not in the Notes. For a fluid particle the mass $\rho\, dV$ is constant and $D\vec{r}/Dt = \vec{v}$ because $\vec{x}_0$ is fixed. The product rule gives

$$\frac{D}{Dt}\left[\vec{r} \wedge \rho\vec{v}\, dV\right] = \left[\vec{v} \wedge \vec{v} + \vec{r} \wedge \frac{D\vec{v}}{Dt}\right]\rho\, dV = \vec{r} \wedge \rho \frac{D\vec{v}}{Dt}\, dV$$

since $\vec{v} \wedge \vec{v} = \vec{0}$. Insert Cauchy's equation (Eq. 3.25), $\rho D\vec{v}/Dt = \nabla \cdot \bar{\bar{\tau}} + \rho\vec{f}_m$, and integrate over the volume:

$$\frac{d}{dt} \int_{V_f} \rho\, \vec{r} \wedge \vec{v}\, dV = \int_{V_f} \vec{r} \wedge (\nabla \cdot \bar{\bar{\tau}})\, dV + \int_{V_f} \rho\, \vec{r} \wedge \vec{f}_m\, dV$$

It remains to show that the first integral on the right equals the surface integral of Eq. 3.37. With the permutation symbol ϵ_{kij}, component k of the integrand is $\epsilon_{kij} r_i\, \partial\tau_{lj}/\partial x_l$. The product rule and $\partial r_i/\partial x_l = \delta_{il}$ give

$$\epsilon_{kij}\, r_i \frac{\partial \tau_{lj}}{\partial x_l} = \frac{\partial}{\partial x_l}\left(\epsilon_{kij} r_i \tau_{lj}\right) - \epsilon_{kij}\, \tau_{ij}$$

The last term vanishes because ϵ_{kij} is antisymmetric in (i, j) and τ_{ij} is symmetric (Eq. 3.19). Gauss' theorem on the first term, together with $\tau_{lj} n_l = \tau_{jl} n_l = (\bar{\bar{\tau}} \cdot \vec{n})_j$, gives

$$\int_V \vec{r} \wedge (\nabla \cdot \bar{\bar{\tau}})\, dV = \int_\Sigma \vec{r} \wedge (\bar{\bar{\tau}} \cdot \vec{n})\, d\sigma$$

which shows that linear momentum together with a symmetric stress tensor is equivalent to Eq. 3.37. The symmetry of the stress tensor is exactly the condition that makes both statements agree. Note that this is not a derivation of Eq. 3.37 from linear momentum alone: in Notes.pdf (pp. 29-30) the symmetry of $\bar{\bar{\tau}}$ (Eq. 3.19) is itself obtained from angular momentum, so the argument would be circular if read that way.

Step 3: Transfer to a moving control volume (Eq. 3.38)

Apply Eq. 3.8 with the density $\vec{\phi} = \rho\, \vec{r} \wedge \vec{v}$, evaluate the right-hand side of Eq. 3.37 on Σ_c, V_c (coincident with Σ_f, V_f at the instant considered) and split the stress with Eq. 3.28. Since $\vec{r} \wedge \left((-p\bar{\bar{I}}) \cdot \vec{n}\right) = -\vec{r} \wedge (p\vec{n})$:

$$\frac{d}{dt}\int_{V_c} \rho\, \vec{r} \wedge \vec{v}\, dV + \int_{\Sigma_c} \rho\, (\vec{r} \wedge \vec{v}) \left[(\vec{v} - \vec{v}_c) \cdot \vec{n}\right] d\sigma = -\int_{\Sigma_c} \vec{r} \wedge (p\vec{n})\, d\sigma + \int_{\Sigma_c} \vec{r} \wedge (\bar{\bar{\tau}}' \cdot \vec{n})\, d\sigma + \int_{V_c} \rho\, \vec{r} \wedge \vec{f}_m\, dV$$

Eq. 3.38

Because $\vec{x}_0$ is fixed, $\vec{r}$ is not differentiated in the control-volume term; the point $\vec{x}_0$ may be placed anywhere, and the choice is a matter of convenience.

Step 4: Uniform ports and the pivot rule (extension)

Extension, not in the Notes. If a port A_k is small compared with its distance to $\vec{x}_0$, then $\vec{r} \approx \vec{r}_k$ is constant over the port. With uniform ρ_k, $\vec{v}_k$ normal to the port and a fixed volume,

$$\int_{A_k} \rho\, (\vec{r} \wedge \vec{v})(\vec{v} \cdot \vec{n})\, d\sigma \approx \vec{r}_k \wedge \Big[\int_{A_k} \rho \vec{v} (\vec{v} \cdot \vec{n})\, d\sigma\Big] = \dot{m}_k\, \vec{r}_k \wedge \vec{v}_k$$

For steady flow the equation reduces to $\sum_k \dot{m}_k\, \vec{r}_k \wedge \vec{v}_k = \sum \vec{M}_{\vec{x}_0}$, with $\dot{m}_k$ negative at inlets. An unknown reaction force $\vec{R}$ applied at the point $\vec{x}_R$ contributes the moment $(\vec{x}_R - \vec{x}_0) \wedge \vec{R}$, which is zero if $\vec{x}_0 = \vec{x}_R$ (or if $\vec{x}_0$ lies on the line of action of $\vec{R}$). This is the pivot rule.

Step 5: Euler turbomachinery equation (every step)

Extension, not in the Notes. Take a fixed annular control volume around the rotor axis z, with an inlet cylinder $r = r_1$ (normal $-\vec{e}_r$) and an outlet cylinder $r = r_2$ (normal $+\vec{e}_r$), with $\vec{x}_0$ on the axis. The flow is periodic and is treated as steady and axisymmetric on average, with ρ, v_r, v_θ uniform on each cylinder. The end walls are impermeable. The z component of the angular momentum per unit mass follows from $\vec{r} = r\vec{e}_r + z\vec{e}_z$ and $\vec{v} = v_r\vec{e}_r + v_\theta\vec{e}_\theta + v_z\vec{e}_z$:

$$(\vec{r} \wedge \vec{v})_z = r\, v_\theta$$

Convective flux in the z component of Eq. 3.38. On the outlet $\vec{v} \cdot \vec{n} = +v_r$ and on the inlet $\vec{v} \cdot \vec{n} = -v_r$. The mass flow through each cylinder is the same, $\dot{m} = \int \rho v_r \, d\sigma$, by Eq. 3.11 in steady flow:

$$\int_{\Sigma_0} \rho\, (r v_\theta)(\vec{v} \cdot \vec{n})\, d\sigma = r_2 v_{\theta 2}\, \dot{m} - r_1 v_{\theta 1}\, \dot{m} = \dot{m}\,(r_2 v_{\theta 2} - r_1 v_{\theta 1})$$

Moments on the right-hand side. A pressure force on a cylinder of constant r is $\mp p\, \vec{e}_r \, d\sigma$, and $(r\vec{e}_r + z\vec{e}_z) \wedge \vec{e}_r = z\, \vec{e}_z \wedge \vec{e}_r = z\, \vec{e}_\theta$, which has no z component. On the end walls the force is along $\pm\vec{e}_z$ and $r\vec{e}_r \wedge \vec{e}_z = -r\vec{e}_\theta$, again with no z component. Viscous stresses on the ports and the moment of the body force about the axis are neglected. The only remaining moment is the moment M_z exerted by the blades (solid surfaces inside the volume) on the fluid:

$$M_z = \dot{m}\,(r_2 v_{\theta 2} - r_1 v_{\theta 1})$$

Here M_z is the torque exerted by the rotor on the fluid. The torque exerted by the fluid on the rotor is $-M_z$ (Newton's third law). Each blade point moves with velocity $\Omega r\, \vec{e}_\theta$, so the power transmitted to the fluid is ΩM_z and the work per unit mass is

$$w_{\text{Euler}} = \frac{\Omega M_z}{\dot{m}} = u_2 v_{\theta 2} - u_1 v_{\theta 1}, \qquad u = \Omega r$$

Dimensional check: $[\dot{m}\, r\, v_\theta] = \mathrm{kg\,s^{-1}\,m\,m\,s^{-1}} = \mathrm{N\,m}$ and $[u\, v_\theta] = \mathrm{m^2/s^2} = \mathrm{J/kg}$.

2. Strategic Pivot Selection Strategy

In many exam problems (such as pipe elbows, nozzle mounts, and rotating sprinklers), several complex linear reaction forces $\vec{R}$ act at physical supports, bearings, or hinges. Solving for these linear reaction forces simultaneously with moments can be tedious and prone to algebraic errors.

THE STRATEGIC PIVOT RULE

Always place the reference reduction point $\vec{x}_0$ **directly at the support hinge, bearing axis, or bolt circle** where unknown linear reaction forces act!

> Because the position vector from $\vec{x}_0$ to the reaction point is zero ($\vec{r} = \vec{x} - \vec{x}_0 = \vec{0}$), the moment arm is zero: $\vec{r} \times \vec{R} = \vec{0}$. The unknown linear reaction forces **completely vanish from the moment balance**, allowing direct solution for holding torque or shaft power in a single line!

1D Streams Approximation

For discrete 1D inlets and outlets with mass flow rates $\dot{m}_i$, position vectors $\vec{r}_i = \vec{x}_i - \vec{x}_0$, and mean velocities $\vec{v}_i$, steady angular momentum reduces to:

$$\sum_j \dot{m}_{\text{out},j}(\vec{r}_{\text{out},j} \times \vec{v}_{\text{out},j}) - \sum_k \dot{m}_{\text{in},k}(\vec{r}_{\text{in},k} \times \vec{v}_{\text{in},k}) = \sum \vec{M}_{\vec{x}_0,\text{ext}}$$

CANONICAL WALKTHROUGH 1 · CL3 & CL15 SETUP

Holding Torque on an Offset Pipe Bend / Mixing Elbow

Physical Setup: Consider a horizontal planar pipe bend in the xy-plane. Water enters at inlet 1 located at $\vec{r}_1 = (0, -L_1, 0)$ flowing in $+y$ direction with speed v_1, area A_1, and gauge pressure p_{g1}. It is deflected through 90° and exits at $\vec{r}_2 = (L_2, 0, 0)$ discharging in $+x$ direction with speed v_2, area A_2, at atmospheric pressure ($p_{g2} = 0$). Mass flow rate is $\dot{m} = \rho A_1 v_1$.

The pipe is anchored to a central mounting hub at the origin $\vec{x}_0 = (0, 0, 0)$. Determine the holding torque $\vec{\mathcal{M}}_0 = \mathcal{M}_z \hat{k}$ that the support must exert to prevent the elbow from spinning.

1 Angular Momentum Fluxes

Compute the convective angular momentum cross-products about $\vec{x}_0 = \vec{0}$:

- **Inlet flux:** Position $\vec{r}_1 = -L_1 \hat{j}$, velocity $\vec{v}_1 = v_1 \hat{j}$.

$$\vec{r}_1 \times \vec{v}_1 = (-L_1 \hat{j}) \times (v_1 \hat{j}) = \vec{0} \qquad \text{(parallel vectors!)}$$

- **Outlet flux:** Position $\vec{r}_2 = L_2 \hat{i}$, velocity $\vec{v}_2 = v_2 \hat{i}$.

$$\vec{r}_2 \times \vec{v}_2 = (L_2 \hat{i}) \times (v_2 \hat{i}) = \vec{0} \qquad \text{(parallel vectors!)}$$

Thus, in this symmetric offset orientation, the convective angular momentum flux is zero! What creates the torque?

2 Moments of External Pressure Forces

Evaluate moments of surface gauge pressure forces:

- **Inlet pressure force:** Pushes inward ($+y$): $\vec{F}_{p1} = p_{g1} A_1 \hat{j}$.

$$\vec{r}_1 \times \vec{F}_{p1} = (-L_1 \hat{j}) \times (p_{g1} A_1 \hat{j}) = \vec{0}$$

- **Outlet pressure force:** Zero because $p_{g2} = 0$ (atmospheric).

General Offset Case: Now suppose the exit is offset by a moment arm d from the x-axis, so $\vec{r}_2 = L_2 \hat{i} + d\hat{j}$.

Then $\vec{r}_2 \times \vec{v}_2 = (L_2 \hat{i} + d\hat{j}) \times (v_2 \hat{i}) = -d v_2 \hat{k}$.

The angular momentum balance yields:

$$\dot{m}(-d v_2 \hat{k}) = \vec{\mathcal{M}}_{\text{anchor}} \implies \vec{\mathcal{M}}_{\text{anchor}} = -\mathbf{\dot{m} d v_2 \hat{k}}$$

The anchor must apply a clockwise holding torque of magnitude $\dot{m} d v_2$ to resist the jet's reaction moment!

CANONICAL WALKTHROUGH 2 · TURBOMACHINERY PRINCIPLE

Euler's Turbomachinery Equation (Rotor Shaft Power)

Problem: A centrifugal compressor or pump rotor rotates at steady angular velocity $\vec{\Omega} = \Omega \hat{e}_z$. Fluid of density ρ enters at inner radius r_1 with mass flow rate $\dot{m}$ and tangential velocity component $v_{\theta 1}$, and exits at outer radius r_2 with tangential velocity component $v_{\theta 2}$. Determine the torque M_z and shaft power $\dot{W}_{\text{shaft}}$ transmitted from the impeller to the fluid.

Derivation Angular Momentum Balance about Axis of Rotation

Select a fixed cylindrical control volume enclosing the rotating impeller. In cylindrical coordinates (r, θ, z), position vectors from the axis are $\vec{r}_1 = r_1 \hat{e}_r$ and $\vec{r}_2 = r_2 \hat{e}_r$.

The cross product of radius with velocity $\vec{v} = v_r\hat{e}_r + v_\theta\hat{e}_\theta + v_z\hat{e}_z$ isolates the tangential velocity component:

$$\vec{r} \times \vec{v} = (r\hat{e}_r) \times (v_r\hat{e}_r + v_\theta\hat{e}_\theta + v_z\hat{e}_z) = rv_\theta\hat{e}_z - rv_z\hat{e}_\theta$$

Projecting onto the axis of rotation z, the angular momentum per unit mass is simply rv_θ.

For steady flow, the net torque M_z exerted by the rotor blades on the fluid equals the net outflow of angular momentum:

$$\mathbf{M_z = \dot{m}(r_2 v_{\theta 2} - r_1 v_{\theta 1})}$$

Multiplying torque by rotor angular speed Ω gives the shaft mechanical power $\dot{W}_{\text{shaft}}$ delivered to the fluid:

$$\dot{W}_{\text{shaft}} = \Omega M_z = \dot{m}(\Omega r_2 v_{\theta 2} - \Omega r_1 v_{\theta 1})$$

Defining blade peripheral speeds $u_1 = \Omega r_1$ and $u_2 = \Omega r_2$, we obtain the celebrated **Euler Turbomachinery Equation**:

$$\mathbf{w_{\text{Euler}} = \frac{\dot{W}_{\text{shaft}}}{\dot{m}} = u_2 v_{\theta 2} - u_1 v_{\theta 1}}$$

PUMPS VS TURBINES SIGN INTERPRETATION

- **Pumps, Fans, and Compressors ($w_{\text{Euler}} > 0$):** The impeller increases the fluid's whirl ($u_2 v_{\theta 2} > u_1 v_{\theta 1}$). Mechanical shaft work is consumed to raise fluid pressure and stagnation enthalpy.
- **Turbines ($w_{\text{Euler}} < 0$):** The fluid enters with high swirl created by stationary guide vanes/nozzles and leaves with reduced tangential velocity ($u_2 v_{\theta 2} < u_1 v_{\theta 1}$). Fluid delivers mechanical shaft work to drive aircraft generators or compressor spools!